**RIP Configuration Lab – Cisco Packet Tracer**

**1. Objective**

To configure Routing Information Protocol (RIP) Version 2 on two routers and enable communication between two different networks.

**2. Devices Used**

- 2 Cisco Routers
- 2 PCs
- Ethernet cables

**3. IP Addressing Table**

| Device | Interface/Network | IP Address | Subnet Mask |
|---|---|---|---|
| R1 | LAN | 192.168.1.1 | 255.255.255.0 |
| R1 | Router Link | 10.0.0.1 | 255.255.255.252 |
| R2 | Router Link | 10.0.0.2 | 255.255.255.252 |
| R2 | LAN | 192.168.2.1 | 255.255.255.0 |
| PC1 | Network Interface | 192.168.1.2 | 255.255.255.0 |
| PC2 | Network Interface | 192.168.2.2 | 255.255.255.0 |

**4. Router R1 Configuration**

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
```

no shutdown

exit

router rip

version 2

no auto-summary

network 192.168.1.0

network 10.0.0.0

end

**5. Router R2 Configuration**

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

exit

router rip

version 2

no auto-summary

network 192.168.2.0

network 10.0.0.0

end

**6. PC Configuration**

- **PC1 IP Address:** 192.168.1.2
- **PC1 Default Gateway:** 192.168.1.1
- **PC2 IP Address:** 192.168.2.2
- **PC2 Default Gateway:** 192.168.2.1
- **Subnet Mask for both PCs:** 255.255.255.0

**7. Verification Commands**

Use the following commands on both routers:

show ip interface brief

show ip route

show ip protocols

The letter R in the routing table indicates a route learned through RIP.

Test connectivity by pinging PC2 from PC1.

**8. Result**

RIP Version 2 was configured on both routers to exchange routing information and enable communication between the two LANs.

**9. Learning Outcomes**

- Learned how to configure RIP Version 2.
- Configured router interfaces with IP addresses.
- Enabled dynamic routing between two routers.
- Verified learned routes using the routing table.
- Tested end-to-end connectivity using ping

**R1**

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
```

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

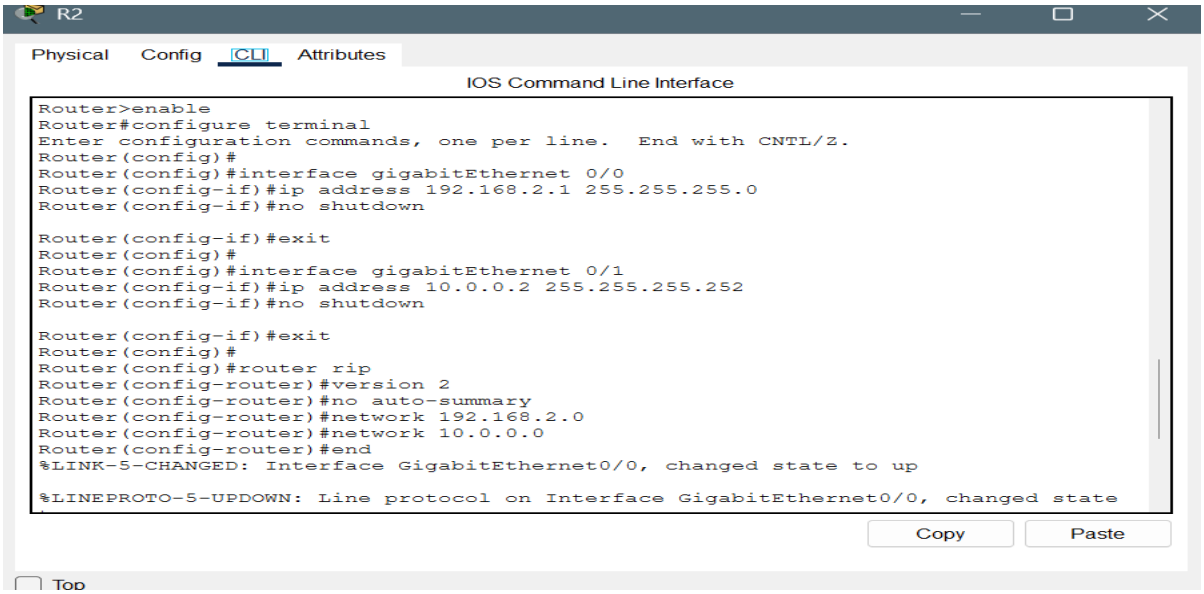
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
```

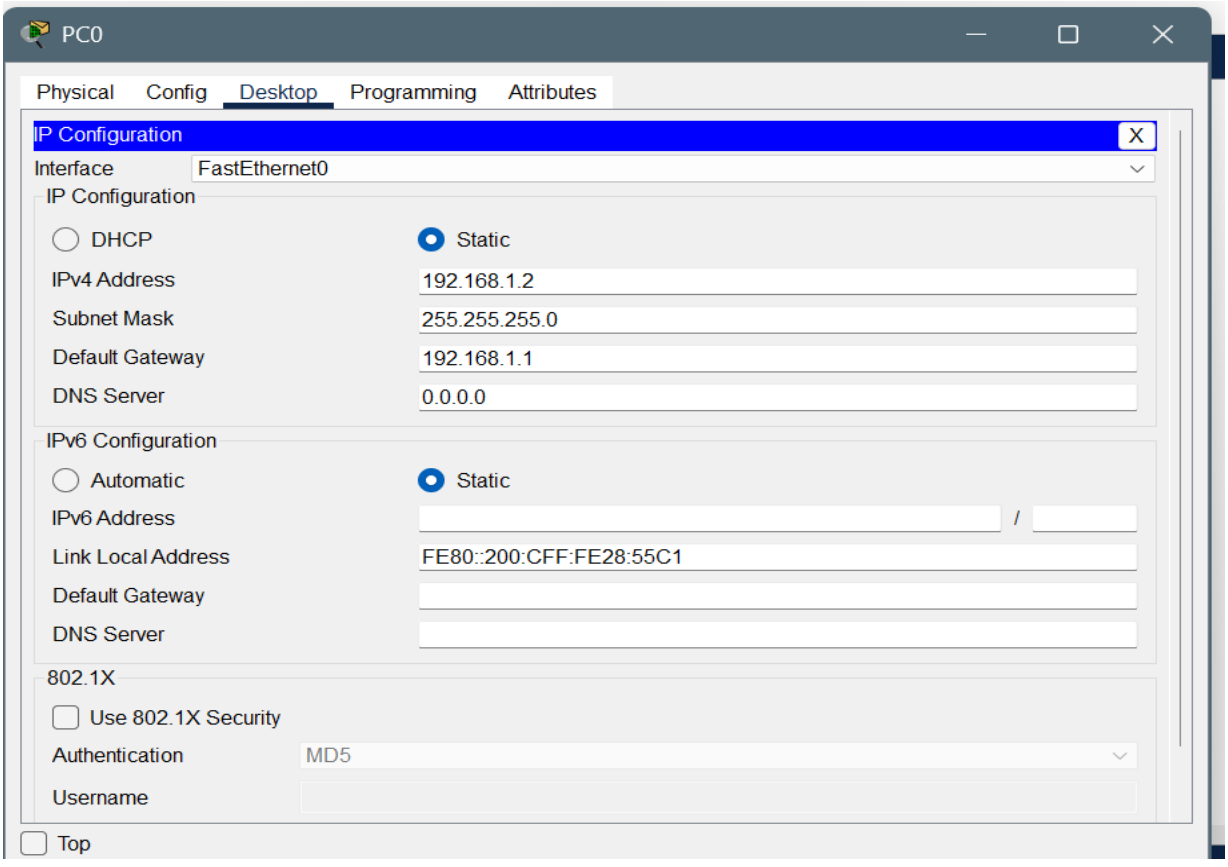
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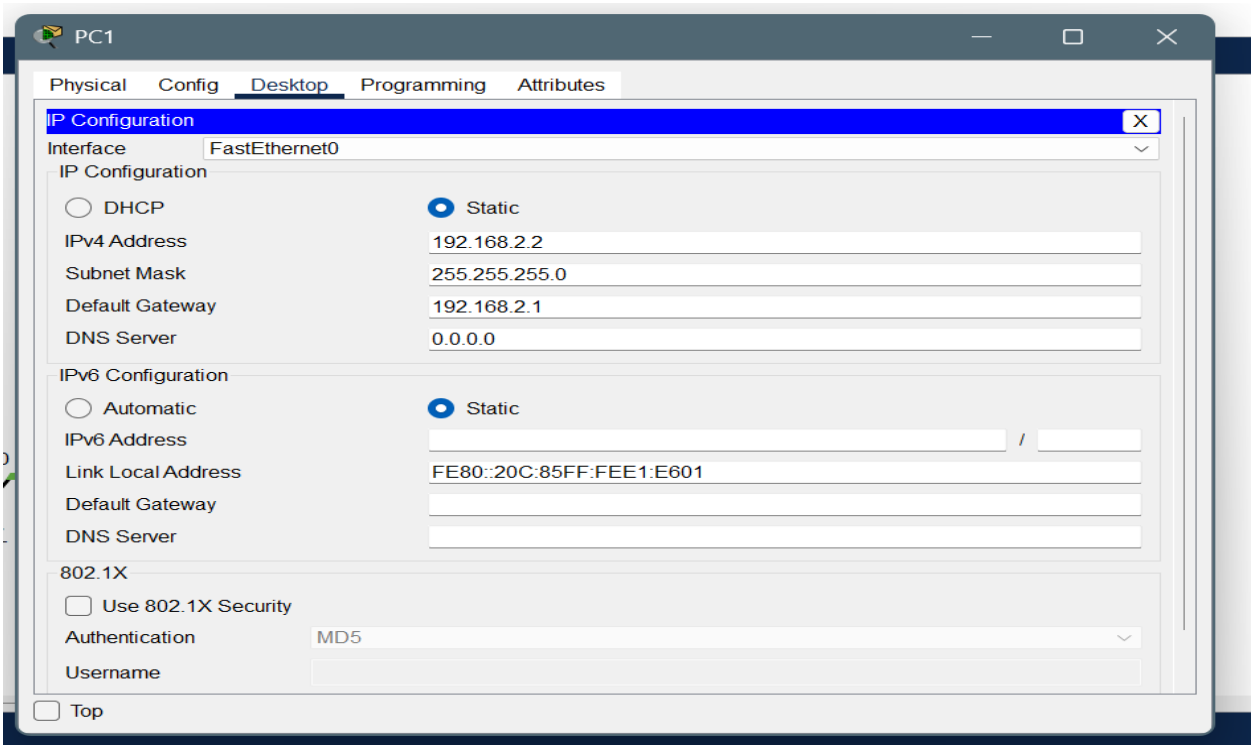
## R2 Configuration



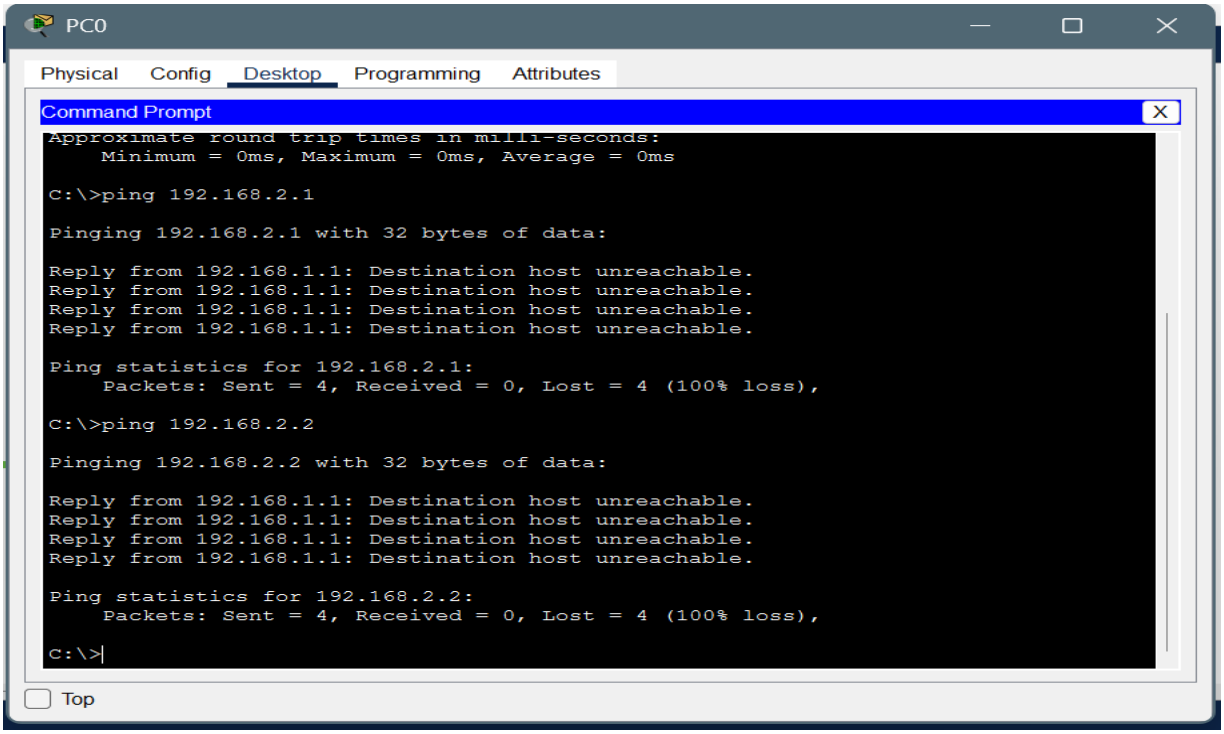

## PC0



**PC1**


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20C:85FF:FEE1:E601
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Top

**Ping**


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
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